(T5) Случайная величина распределена равномерно на отрезке $[\theta; 2\theta]$

$\xi \sim R[\theta; 2\theta]$ $\quad p(x) = \frac{1}{\theta}\{[\theta; 2\theta]\}$

а) x_n - выборка, найти оценки параметра θ методом моментов и методом максимального правдоподобия.

ОММ:

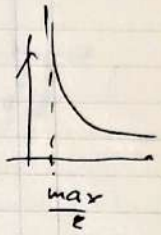
$$\alpha_1 = \int_{-\infty}^{\infty} x \frac{1}{\theta} dx = \int_{\theta}^{2\theta} \frac{x}{\theta} dx = \frac{x^2}{2\theta}\Big|_{\theta}^{2\theta} = \frac{3}{2}\theta = \bar{x}$$

$$\hookrightarrow \tilde{\theta}_1 = \frac{2}{3}\bar{x}$$

ОМП:

$$L(\theta) = \prod_{i=1}^{n} p(x_i; \theta) = \frac{1}{\theta^n}\{\theta < x_i \leq 2\theta\} =$$

$$= \frac{1}{\theta^n}\left\{\frac{\max x_i}{2} \leq \theta \leq \min x_i\right\}$$



$$\sup_{(0;+\infty)} L(\theta) \longrightarrow \tilde{\theta}_2 = \frac{\max x_i}{2} = \frac{x_{max}}{2}$$

б) Проверить оценки на несмещенность и состоятельность.

1) $$M\tilde{\theta}_1 = \frac{2}{3}M\bar{x} = \frac{2}{3}M\frac{1}{n}\sum_{k=1}^{n} x_k = \frac{2}{3}\cdot\frac{1}{n}\cdot n\cdot M\xi =$$

$$= \frac{2}{3}\cdot\int_{-\infty}^{\infty} x\,dF = \frac{2}{3}\int_{\theta}^{2\theta} x\frac{1}{\theta}dx = \frac{2}{3}\cdot\frac{3}{2}\theta = \theta \Rightarrow \text{несмещ.}$$

Проверим сост. по дост. усл:

$$D\tilde{\theta}_1 = D\left(\frac{2}{3}\bar{x}\right) = \frac{4}{9}D\bar{x} = \frac{4}{9}\cdot\frac{1}{n^2}\cdot n D\xi = \frac{4}{9n}\cdot$$

$$\cdot\left(\int_{-\infty}^{\infty} x^2 dF - (M\xi)^2\right) = \frac{4}{9n}\left(\int_{\theta}^{2\theta}\frac{x^2}{\theta}dx - \left(\frac{3}{2}\theta\right)^2\right) = \frac{4}{9n}\left(\frac{1}{\theta}\left(\frac{8\theta^3}{3} - \frac{\theta^3}{3}\right)\right.$$

$$\left. - \frac{9}{4}\theta^2\right) = \frac{4}{9n}\cdot\frac{1}{12}\theta^2 \xrightarrow{n\to\infty} 0 \Rightarrow \text{состоят.}$$

2) $M\tilde{\theta}_2 = \frac{1}{2} M x_{max} = \frac{1}{2}\int_{\theta}^{2\theta} x \cdot \frac{n}{\theta}\left(\frac{x-\theta}{\theta}\right)^{n-1} dx =$

$= \Big/ \frac{x-\theta}{\theta} = t \Big/ = \frac{1}{2}\int_0^1 (\theta t + \theta)\frac{n\theta}{\theta} t^{n-1} dt = \frac{1}{2}\int_0^1 n(\theta t^n + \theta t^{n-1}) dt$

$= \frac{1}{2} n\theta \left(\frac{t^{n+1}}{n+1} + \frac{t^n}{n}\right)\Big|_0^1 = \frac{1}{2} n\theta\left(\frac{1}{n+1} + \frac{1}{n}\right) = \frac{1}{2}\theta \frac{n+n+1}{n+1} =$

$= \theta \frac{2n+1}{2n+2} \neq \theta \Rightarrow$ смещенная

Исправим: $\tilde{\theta}_2' = \frac{2n+2}{2n+1} \frac{2x_{max}}{2} = \frac{n+1}{2n+1} 2x_{max}$ – несмещ

Проверим состоятельность исправленной оценки по дост. зам:

$D\tilde{\theta}_2' = \left(\frac{n+1}{2n+1}\right)^2 D x_{max}$

$Dx_{max} = Mx_{max}^2 - (Mx_{max})^2$

$Mx_{max}^2 = \int_{\theta}^{2\theta} x^2 \frac{n}{\theta}\left(\frac{x-\theta}{\theta}\right)^{n-1} dt = \Big/ \frac{x-\theta}{\theta} = t \Big/ =$

$= \int_0^1 (\theta t + \theta)^2 n t^{n-1} \frac{\theta}{\theta} dt = n\theta^2 \int_0^1 (t^{n+1} + 2t^n + t^{n-1}) dt$

$= n\theta^2\left(\frac{1}{n+2} + \frac{2}{n+1} + \frac{1}{n}\right) = \frac{\theta^2(n^2+n+2n^2+4n+n+2)}{(n+2)(n+1)}$

$= \theta^2 \cdot 2\frac{2n^2+4n+1}{n^2+3n+2}$

$\Rightarrow D\theta_2' = \left(\frac{n+1}{2n+1}\right)^2\left(2\theta^2 \cdot \frac{2n^2+4n+1}{n^2+3n+2} - \theta^2\left(\frac{2n+1}{n+1}\right)^2\right) =$

$= \theta^2 \frac{(4n^2+8n+2)(n+1) - (4n^2+4n+1)(n+2)}{(n+2)(2n+1)^2} =$

$= \theta^2 \frac{4n^3+8n^2+2n+4n^2+8n+2 - 4n^3-4n^2-n-8n^2-8n-2}{(n+2)(2n+1)^2} =$

$= \theta^2 \cdot \frac{n}{(n+2)(2n+1)^2} \xrightarrow[n\to\infty]{} 0 \Rightarrow$ сост

c) Сравнить эффективность исправленных оценок

$\tilde{\theta}_1$ и $\tilde{\theta}_2'$ – несмещ., сост

сравним $D\tilde{\Theta}_1$ и $D\tilde{\Theta}_2'$

$$\frac{\Theta^2}{27n} \quad \text{и} \quad \frac{n\Theta^2}{(2n+1)^2(n+2)}$$

$$(2n+1)^2(n+2) > 27n^2$$

выр. n (наверное 3)

$\hookrightarrow$ $\tilde{\Theta}_2'$ эффективное для $n \geq 3$

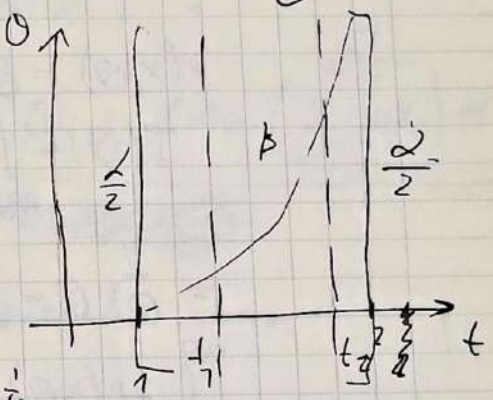
d) Построить точечный доверительный интервал для Θ.

$$f(\overline{x_n};\Theta) = \frac{x_{max}}{\Theta} \sim n(t-1)^{n-1}\{[1,2]\}$$

$$F(t) = P\left(\frac{x_{max}}{\Theta} \leq t\right) = P(x_{max} \leq t\Theta) = \left(\frac{t\Theta - \Theta}{\Theta}\right)^n =$$

$$= (t-1)^n$$

$$\hookrightarrow \rho = n(t-1)^{n-1}$$



$$t_1 = \rho\,\frac{1-\beta}{2} \qquad t_2 = \rho\,\frac{1+\beta}{2}$$

$$\hookrightarrow F(t_1) = \frac{1-\beta}{2} \qquad F(t_2) = \frac{1+\beta}{2}$$

$$t_1 = 1 + \left(\frac{1-\beta}{2}\right)^{\frac{1}{n}} \hookrightarrow t_2 = 1 + \left(\frac{1+\beta}{2}\right)^{\frac{1}{n}}$$

$$P\left(t_1 < \frac{x_{max}}{\Theta} < t_2\right) = \beta$$

$$\hookrightarrow \frac{x_{max}}{t_2} < \Theta < \frac{x_{max}}{t_1}$$

e) Построить ассимптотический доверительный интервал для параметра Θ:

1) $\hat{\Theta}_1 = \frac{2}{3}\bar{x} = g(\tilde{\alpha})$ (ОММ)

$$\sqrt{n}\,\frac{g(\tilde{\alpha}) - g(\alpha)}{\sigma(\tilde{\alpha})} \rightsquigarrow N(0,1)$$

$$\nabla g(\alpha) = \frac{2}{3}$$

$$k_{11} = \alpha_{1+1} - \alpha_1\alpha_1$$

$$\sigma^2(\alpha) = \nabla g^T(\alpha)\, k\, \nabla g(\alpha) = \frac{2}{3}(\alpha_2 - \alpha_1^2)\cdot\frac{2}{3} = \frac{4}{9}(\alpha_2 - \alpha_1^2)$$

т.о. $\sqrt{n}\,\dfrac{\widetilde{\Theta}_1 - \Theta}{\frac{2}{3}\sqrt{\overline{\alpha}_2 - \overline{\alpha}_1^2}} \to N(0;1)$

$$\overline{\alpha}_2 = \frac{1}{n}\sum_{i=1}^{n} x_i^2$$

$$t_1 = U_{\frac{1-\beta}{2}} \qquad t_2 = U_{\frac{1-\beta}{2}}$$

$$\hookrightarrow U_{\frac{1-\beta}{2}} < \sqrt{n}\,\frac{\widetilde{\Theta}_1 - \Theta}{\frac{2}{3}\sqrt{\overline{\alpha}_2 - \overline{\alpha}_1^2}} < U_{\frac{1+\beta}{2}}$$

$$\frac{2}{3\sqrt{n}}\, t_1 \sqrt{\overline{\alpha}_2 - \overline{\alpha}_1^2} < \widetilde{\Theta}_1 - \Theta < \frac{2}{3\sqrt{n}} \sqrt{\overline{\alpha}_2 - \overline{\alpha}_1^2}\, t_2$$

$$\widetilde{\Theta}_1 - \frac{2\sqrt{\overline{\alpha}_2 - \overline{\alpha}_1^2}}{3\sqrt{n}}\, t_2 < \Theta < \widetilde{\Theta}_1 - \frac{2}{3}\frac{\sqrt{\overline{\alpha}_2 - \overline{\alpha}_1^2}}{\sqrt{n}}\, t_1$$

2) $\widetilde{\Theta}_2' = \dfrac{n+1}{2n+1} X_{max}$

$p(x;\Theta) = \frac{1}{\Theta}\{(\Theta; 2\Theta)\}$ – непр дифф. при $\Theta > 0$

$$I(\Theta) = \int_{A(\Theta)} \left(\frac{\partial \ln p(x;\Theta)}{\partial \Theta}\right)^2 p(x;\Theta)\,dx = \int_{\Theta}^{2\Theta} \frac{1}{\Theta^2}\,\frac{1}{\Theta}\,dx =$$

$= \dfrac{1}{\Theta^3}\,\Theta = \dfrac{1}{\Theta^2}$ непр. и > 0

$$\int_{\Theta}^{2\Theta} \frac{\partial p(x;\Theta)}{\partial \Theta}\,dx = \int_{\Theta}^{2\Theta} -\frac{1}{\Theta^2}\,dx = -\frac{1}{\Theta} \neq 0$$

$\hookrightarrow$ Оценка не регулярна, нельзя асимптотич. оценить.

(16) Случайная величина имеет распределение Парето:

$$p(x) = \begin{cases} \frac{\theta - 1}{x^{\theta}}, & x \geq 1 \\ 0, & x < 1 \end{cases} \qquad \theta > 1$$

а) По выборке объема n найти оценку параметра θ методом максимального правдоподобия

$$L(\theta) = \prod_{i=1}^{n} p(x_i;\theta) = \frac{(\theta-1)^n}{\left(\prod_{i=1}^{n} x_i\right)^{\theta}} \quad \{x_{(1)} \geq 1\}$$

$$(\ln L)' = \left(n \ln(\theta-1) - \theta \sum_{i=1}^{n} \ln x_i\right)' =$$

$$= \frac{n}{\theta-1} - \sum_{i=1}^{n} \ln x_i = 0 \Rightarrow \hat{\theta} = 1 + n\left(\sum_{i=1}^{n} \ln x_i\right)^{-1}$$

$$(\ln \theta)'' = -\frac{n}{(\theta-1)^2} < 0$$

$\hookrightarrow$ это действительно max.

б) Построить доверительный интервал для медианы.

Регулярна ли модель?

$p(x;\theta)$ непр. диф по θ на $\theta > 1$

$$\frac{\partial p}{\partial \theta} = \frac{1 - \ln x(\theta-1)}{x^{\theta}}$$

$$\int_{-\infty}^{\infty} \frac{\partial p}{\partial \theta} dx = \int_{-\infty}^{\infty} x^{-\theta}(1 - \ln x(\theta-1))dx = \int_{-\infty}^{\infty} x^{-\theta+1}\Big|_{1}^{\infty} -$$

$$-(\theta-1)\Big|_{1}^{\infty} x^{-\theta} \ln x\, dx = \frac{1}{1-\theta} - (\theta-1) \cdot \frac{-1}{(\theta-1)^2} = 0$$

$$\Rightarrow \int_{-\infty}^{\infty} \frac{\partial p}{\partial \theta} dx = \frac{\partial}{\partial \theta} \int_{-\infty}^{\infty} p(x)dx$$

$\ln p = \ln(\theta-1) - \theta \ln x$

$$\frac{\partial \ln p}{\partial \theta} = \frac{1}{\theta-1} - \ln x$$

$$I(\theta) = \int_1^{+\infty} \frac{\theta-1}{x^\theta}\left(\frac{1}{\theta-1} - \ln x\right)^2 dx =$$

$$\cancel{= \int_1^{+\infty} \frac{\theta-1}{x^\theta}\left(\frac{1}{(\theta-1)^2} - 2\frac{\ln x}{\theta-1} + \ln^2 x\right) dx =}$$

$$\cancel{= \int_1^{+\infty} \frac{dx}{x^\theta(\theta-1)} - 2\int_1^{+\infty} \frac{\ln x}{x^\theta} dx + \int_1^{+\infty} \frac{(\theta-1)\ln^2 x}{x^\theta} dx =}$$

$$= \Big/ y = \ln x \Big/ = \int_0^{+\infty} (\theta-1) e^{-\theta y}\left(\frac{1}{\theta-1} - y\right)^2 e^y dy$$

$$= \int_0^{+\infty} (\theta-1) e^{y(1-\theta)}\left(\frac{1}{\theta-1} - y\right)^2 dy = \Big/ t = \theta-1 \Big/ =$$

$$= \int_0^{+\infty} t e^{-ty}\left(\frac{1}{t} - y\right)^2 dy = \int_0^{+\infty} e^{-ty}\frac{1}{t} dy - 2\int_0^{+\infty} e^{-ty} y\, dy$$

$$+ t\int_0^{+\infty} e^{-ty} y^2 dy = \frac{1}{t^2} - 2\frac{1}{t^2} + \frac{2}{t^2} = \frac{1}{t^2} =$$

$= \frac{1}{(\theta-1)^2}$ — непр. и >0 на Θ $\Rightarrow$ модель регулярна

· $p(x;\theta)$ — $n>2$ раз непр. диф. по θ на Θ

· $\frac{\partial^2 p}{\partial \theta} = \ln x \cdot x^{-\theta}(\theta \ln x - \ln x - 2)$

$$\int_1^{+\infty} \frac{\partial^2 p}{\partial \theta^2} dx = \int_1^{+\infty} (\ln^2 x \cdot x^{-\theta}(\theta-1) - 2\ln x \cdot x^{-\theta}) dx = 0$$

$\longrightarrow$ состоят., асимпт. нормальн, асимпт. эффективна и асимпт. нормальна

Медиана: $F(x_{\frac{1}{2}}) = \frac{1}{2}$

$$\int_1^{x_{\frac{1}{2}}} \frac{\theta-1}{x^{\theta}}\,dx = -\frac{1}{x_{\frac{1}{2}}^{\theta-1}} + 1 = \frac{1}{2}$$

откуда $x_{\frac{1}{2}} = 2^{\frac{1}{\theta-1}}$

$$g(\hat{\theta}) = \sqrt[\hat{\theta}-1]{2}$$

$$\sqrt{n}\,\frac{g(\hat{\theta}) - g(\theta)}{\sqrt{\nabla^T g(\hat{\theta})\, I^{-1}(\theta)\, \nabla g(\theta)}} \rightsquigarrow N(0,1)$$

$$\nabla g(\hat{\theta}) = 2^{\frac{1}{\theta^2-1}} \ln 2 \left(-\frac{1}{(\hat{\theta}-1)^2}\right)$$

$$\hookrightarrow \sqrt{n}\,\frac{(g(\hat{\theta}) - g(\theta))}{2^{\frac{1}{\theta-1}} \ln 2}\,(\theta^2-1) \rightsquigarrow N(0,1)$$

$$\longrightarrow P\left(g(\hat{\theta}) - \frac{1{,}96 \ln 2 \cdot 2^{\frac{1}{\theta-1}}}{\sqrt{n}\,(\hat{\theta}-1)} < x_{\frac{1}{2}} < g(\hat{\theta}) + \frac{1{,}96 \ln 2 \cdot 2^{\frac{1}{\theta-1}}}{\sqrt{n}\,(\hat{\theta}-1)}\right) = \beta$$

$$g(\hat{\theta}) - \frac{1{,}96 \ln 2 \cdot 2^{\frac{1}{\theta-1}}}{\sqrt{n}\,(\hat{\theta}-1)} < x_{\frac{1}{2}} < g(\hat{\theta}) + \frac{1{,}96 \ln 2 \cdot 2^{\frac{1}{\theta-1}}}{\sqrt{n}\,(\hat{\theta}-1)}$$

в) Построить асимптотич. довер. интервал для θ:

$$\frac{\sqrt{n}\,(\hat{\theta} - \theta)}{\hat{\theta} - 1} \rightsquigarrow N(0,1)$$

$$-1{,}96 < \sqrt{n} \cdot \frac{1}{\sqrt{n}} \left(\sum_{i=1}^{n} \ln x_i\right) \left(1 + n\left(\sum_{i=1}^{n} \ln x_i\right)^{-1} - \theta\right) < 1{,}96$$

$$\hookrightarrow 1 - \frac{1{,}96 - \sqrt{n}}{\sqrt{n} \cdot \frac{1}{n}\sum_{i=1}^{n} \ln x_i} < \theta < 1 + \frac{1{,}96 + \sqrt{n}}{\sqrt{n}\,\frac{1}{n}\sum_{i=1}^{n} \ln x_i}$$